

Unifying Bayesian Estimation and Markov Chains for Forecast Models

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1 Bayesian Probability Estimation

1.1 Prior Specification

Let p be the true probability of upward movement. We adopt a conjugate Beta prior:

$$p \sim \text{Beta}(a_0, b_0) \propto p^{a_0-1}(1-p)^{b_0-1} \quad (1)$$

For Laplace's uniform prior: $a_0 = 1, b_0 = 1$.

1.2 Likelihood Function

Given n Bernoulli trials with k successes:

$$\mathcal{L}(p|k, n) = \prod_{i=1}^n p^{X_i}(1-p)^{1-X_i} = p^k(1-p)^{n-k} \quad (2)$$

1.3 Posterior Derivation

Using Bayes' theorem:

$$f(p|k, n) = \frac{\mathcal{L}(p|k, n)\pi(p)}{\int_0^1 \mathcal{L}(p|k, n)\pi(p)dp} \quad (3)$$

$$= \frac{p^k(1-p)^{n-k}p^{a_0-1}(1-p)^{b_0-1}}{B(k+a_0, n-k+b_0)} \quad (4)$$

$$= \text{Beta}(p|k+a_0, n-k+b_0) \quad (5)$$

Where the normalizing constant is the Beta function:

$$B(a, b) = \int_0^1 t^{a-1}(1-t)^{b-1}dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad (6)$$

1.4 Posterior Expectation

The Laplace-rule estimator becomes:

$$\hat{p} = \mathbb{E}[p|k, n] = \frac{k + a_0}{n + a_0 + b_0} = \frac{k + 1}{n + 2} \quad (7)$$

2 Markovian Price Dynamics

2.1 State Space Definition

- Discrete price states: $\mathcal{S} = \{s_1, \dots, s_m\}$
- State vector: $\mathbf{x}_t = [\mathbb{P}(X_t = s_i)]_{i=1}^m \in \Delta^{m-1}$

2.2 Transition Matrix Formulation

The generalized transition matrix combines prior beliefs $\mathbf{Q}$ and data-driven dynamics:

$$\mathbf{S}_t = \alpha \mathbf{Q} + (1 - \alpha) [\hat{p}_t \mathbf{S}^+ + (1 - \hat{p}_t) \mathbf{S}^-] \quad (8)$$

Where:

- $\mathbf{Q} \in \mathbb{R}^{m \times m}$: Prior transition matrix (e.g., uniform transitions)
- $\mathbf{S}^+$: Upward transition operator with reflecting boundary:

$$S_{ij}^+ = \begin{cases} 1 & j = \min(i + 1, m) \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

- $\mathbf{S}^-$: Downward transition operator:

$$S_{ij}^- = \begin{cases} 1 & j = \max(i - 1, 1) \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

2.3 State Evolution

The probabilistic state update follows:

$$\mathbf{x}_{t+1} = \mathbf{x}_t \mathbf{S}_t = \alpha \mathbf{x}_t \mathbf{Q} + (1 - \alpha) [\hat{p}_t \mathbf{x}_t \mathbf{S}^+ + (1 - \hat{p}_t) \mathbf{x}_t \mathbf{S}^-] \quad (11)$$

3 Convergence Properties

[Ergodic Convergence] For $\alpha \in (0, 1)$ and irreducible $\mathbf{Q}$, the Markov chain with transition matrix $\mathbf{S}_t$:

1. Is geometrically ergodic

2. Has unique stationary distribution π satisfying:

$$\pi = \pi [\alpha \mathbf{Q} + (1 - \alpha) \mathbb{E}[\hat{p}_\infty \mathbf{S}^+ + (1 - \hat{p}_\infty) \mathbf{S}^-]] \quad (12)$$

3. Exhibits exponential convergence:

$$\|\mathbf{x}_t - \pi\|_1 \leq C\rho^t, \quad \rho < 1 \quad (13)$$

Step 1: Irreducibility

Since $\mathbf{Q}$ is irreducible and $\alpha > 0$, all states communicate through the convex combination of transitions.

Step 2: Aperiodicity

The self-transition probability from $\mathbf{Q}$ ensures:

$$\gcd\{t : (\mathbf{S}_t^n)_{ii} > 0\} = 1 \quad (14)$$

Step 3: Geometric Ergodicity

Follows from Theorem 15.0.1 in Meyn-Tweedie (1993) for Doeblin chains. The minorization condition holds as $\mathbf{S}_t \geq \alpha \mathbf{Q}$.

4 Computational Analysis

4.1 Algorithmic Complexity

- **Time Complexity:**

$$O(T(m^2 + c)) \quad \text{where } c = \text{Bayesian update cost} \quad (15)$$

- **Space Complexity:**

$$O(m^2) \quad (\text{Storing transition matrices}) \quad (16)$$

4.2 Performance Characteristics

Operation	Naive	Optimized
Matrix construction	$O(m^2)$	$O(m)$ (sparse)
State update	$O(m^2)$	$O(m)$ (shift ops)
Bayesian learning	$O(1)$	$O(1)$

4.3 Pros and Cons

- **Advantages:**

- Theoretical Guarantees: Provable convergence
- Adaptivity: Learns regime changes via $\hat{p}_t$
- Numerical Stability: Convex combinations prevent overflow

- **Limitations:**

- Curse of Dimensionality: $O(m^2)$ scaling for dense matrices
- Parameter Sensitivity: Requires careful tuning of (α, λ)
- Discretization Error: From finite state approximation

5 Empirical Validation

5.1 Convergence Diagnostics

The spectral gap $\gamma = 1 - \|\lambda_2\|$ determines mixing time:

$$t_{\text{mix}}(\epsilon) \leq \frac{1}{\gamma} \log \left(\frac{1}{\epsilon} \right) \quad (17)$$

Where λ_2 is the second largest eigenvalue of $\mathbf{S}_t$.

5.2 Heatmap Generation

The price probability heatmap visualizes:

$$\mathbf{H} = [\mathbf{x}_1^\top; \dots; \mathbf{x}_T^\top] \in \mathbb{R}^{T \times m} \quad (18)$$

With columns as price states and rows as time steps.